Visualizing Modeled Social Networks: a Provocation *

Anonymized for Review

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Abstract: Network scholarship regularly leverages visualizations. They are thought to enhance readers' recall, interpretation, and understanding of scholars' intended takeaways. However, visualizations combine multiple competing demands that do not readily translate into a single optimal strategy for realizing those benefits. Here, I distinguish between the scientific (1) purposes and (2) principles of network visualization, which are then differentially used across (3) stages of the research process. Drawing on these non-overlapping demands, I encourage the field to consider whether we can improve the utility of visualizations for certain combinations of those aims by presenting *modeled* results in network visualizations.

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*... All code for replication is available at [removed for review].

Social network research relies heavily on data visualization (Freeman 2004). These visualizations can serve to characterize, simplify, and focus empirical and/or theoretical claims (Moody et al. 2005; Rawlings et al. 2023). Visualizations are generally more readily digestible than tables of (seemingly endless) coefficients, significance tests, and summary statistics (Healy 2018; Schwabish 2021). Additionally, with the increasing ubiquity of network verbiage in society, they can even convey ideas in ways that require less "translation" work for audiences from outside of academic and research niches (Sayama et al. 2016; Wolff et al. 2026). Quite simply, network visualization is not only common, but can be incredibly useful (Birkett et al. 2021).[1]

Traditionally, this has often taken the form of directly visualizing instance(s) of empirically **observed** social networks–whether snapshots of single data waves, or evolving network "movies" for dynamic relational data patterns (Ognyanova 2025).[2] That is, such visualizations are intended to characterize the data *per se*. While this practice predominates, it's worth asking: (1) what purposes motivate how and why we visualize, and (2) what additional principles govern appropriate strategies for accomplishing those aims, then (3) whether (there are conditions for which) an alternative strategy serves those *purposes* and *principles* better than the existing approach?

In this paper, I suggest that an orientation that assumes a *single* visualization orientation can successfully address *multiple* sets of aims under-delivers on the potential value of network visualization–as often arises when goals compete (Pache and Santos 2010). In particular, my primary contention is that certain uses of social network visualizations would be better served by visualizing modeled networks rather than observed network data, especially for research's theoretical implications. To elaborate these differences, I describe how the aims across the *stages* of the research process often engage the outlined purposes and principles in distinct combinations. After making this case, I illustrate the shift to modeled network visualizations in three ways–a trivial example common to how many of us already teach social network modeling, aligning this shift with several parallel existing practices, and an applied example from a recent research paper.

Before continuing, let me explicitly note that I use "model" in an intentionally *theoretical*, and not a *statistical* sense. I avoid relegating this point to a footnote to highlight that the strategy I encourage does not apply only to statistically modeled results–instead being the elaboration of the theoretical interpretive takeaways that arise from *any* systematic empirical analytic strategy. Moreover, also note that this broader applicability reflects that in some ways, what I suggest *imports* practices from non-statistical orientations to network research as much as (or perhaps even more so) than developing a method for statistical applications that can

[1] As with any data presentation strategy–visual or otherwise–there are also ample examples of how to do this poorly (e.g., what have come to be known as unintelligible network "hairballs"); I have a few published visuals of my own that I'd *substantially* rework if given the opportunity today.

[2] These typically occur *after* data pre-processing has taken place (e.g., cleaning, recoding, etc.), but no further data reduction steps (i.e., are not the product of data *summaries* or analytic interpretations).

be exported to other use cases.[3]

Why (network) visualization?

So, what are the benefits of data visualizations generally, and how do those apply specifically in networks research? Here, I begin from the assumption that SNA visualizations seek to optimize their *scientific utility* (broadly construed), and highlight some shortcomings that have emerged in our common practices, which have weakened our ability to accomplish those aims–particularly by assuming a single visualization strategy can address multiple (sometimes competing) sets of aims. Moreover visualization may aid in constructing or interpreting analytic approaches, but is not itself strictly analytic–that is, visualizations are communicative tools, not methodological ones. This is especially important for noting that visualizations provide and convey assertions, rather than means for testing them; thus I do *not* address how to conduct and optimize analyses here. Those are separate tasks that I assume visualization complements, rather than overlaps with or replaces.

Many of the benefits for network scholarship mirror those that motivate scientific data visualization more broadly (Rawlings et al. 2023). A fundamental premise is that data visualization enhances **memory** (Schwabish 2021); this reflects what we know from a number of old adages–including that there's simply value in beauty (Tufte 2006; Halpern 2015). Moreover, this memory benefit leverages the potential **efficiency** available in data visualization that simply cannot be achieved in other strategies for presenting evidence (Healy 2018).[4] Edward Tufte has prioritized not only using this benefit, but attempting to find ways to optimize it, e.g., via efforts to maximize what he called the "ink-to-information" ratio (Tufte 1983). These memory-oriented enhancements can arise both via improved encoding processes *and* recall–e.g., the simple act of including visualizations leverages the complementary of multi-modal communication strategies' differential strengths and weaknesses (e.g., compared to text Sweller (1988)].

The data we use to represent social processes are frequently high-dimensional (especially networks), and visualizations provide a means for **simplifying** that complexity in structured, interpretable formats (Wickham et al. 2023).[5] Simplification enhances visualizations communicative potential, while also enhancing readers' ability to understand and interpret complex social phenomenon (Schwabish 2021). While statistics can summarize data in principled ways, data visualization rarely simply mimics these other forms and representations of data reduction. Instead, they are also "attention-drawing" efforts that highlight certain features of the data,

[3]Specifically, some of the elements proposed below are directly adopted from other corners of SNA where they are already variably common practices (if labeled differently), and (partially) pave the way for how statistically oriented researchers can implement the approach. To this end, the examples below draw on a range of analytic orientations that are not exclusively formal inferential statistical modeling approaches.

[4]There are many pithy statements about the vast amounts of information that can be conveyed in a single image.

[5]Some contend that this is a *constraint* of data visualization imposed on data presentation strategies (by obfuscating its richness), rather than its benefit (Healy and Moody 2014).

while simultaneously deprioritizing others (Leifer 1992).

So, how do network scholars in particular prioritize these aims? Network scholars especially leverage these dimension reduction approaches by translating summaries of (theoretically) salient *social space* into the approaches for producing visualized representations (Rawlings et al. 2023). One such "attention-drawing" move a research may use would highlight certain relational patterns (e.g., community structure), while deprioritizing others (e.g., homophily). For example, to emphasize the changes in political polarization over time in the US Congress, Moody and Mucha (2013) employ increased coarse-graining to highlight *group* size and positioning over time, and juxtapose that against the decreased *individual* positioning and distancing.

Paralleling the myriad dimensions of analytically capturing empirical network properties, network visualizations also require a variety of strategies to represent these patterns of social structure. Below, I outline ways that network visualizations have not optimized this varied potential; and my hope is tor encourage, more, more-varied, and more tailored uses of visualizations in ways that will afford each instance a greater likelihood of achieving the researchers' particular needs. As a starting point, network research could benefit from rethinking optimal visualization strategies depending on their exploratory (Wickham et al. 2023), conceptual/theoretical (Brady et al. 2020), or explanatory aims (Healy and Moody 2014).

Multidimensional aims: scientific purposes and principles, across research stages

To guide a more tailored approach to optimizing network visualizations, let's begin by delineating: the differences among multidimensional sets of aims, how they compete with and complement one another, and how researcher practices combine across these dimensions. First, I distinguish several distinct scientific **purposes**—the communicative reasons for employing visualization. Second, I identify additional scientific **principles** that govern visualization practices in pursuing those purposes. Third, I differentiate three **stages** of research in which visualizations can prove useful: exploration, conceptualization, and interpretation. Together, these dimensions outline how particular visualization practices necessarily prioritize different **combinations** of these purposes and principles as their roles shift across the stages of the research process. I draw on the broader literature on data visualizations to elaborate these sets of distinctions, and highlight any network-specific considerations across them.

Scientific purposes of network visualization

A first dimension in leveraging the utility of network visualizations concerns the various purposes they may accomplish. Although these purposes principally orient visualization toward distinct goals, they are not fully mutually exclusive (Healy and Moody 2014).

Description seeks to systematically characterize patterns, such as general tendencies or variation, to capture the state of things as they are as accurately as possible (Schwabish 2021). In networks, this can span multiple levels—including features of nodes, ties, sections within networks, or networks as a whole (Rawlings et al. 2023; Moody et al. 2005).

Case interpretation makes sense of the particularities of a case, which can help identify its important features as contrasted against the alternatives that could have been (Ragin and Becker 1992; Collins et al. 2024). Additionally, this provides a lens through which to assess how broadly those observed features should inform our understanding of the wider social setting the case represents (Ayala and Koch 2019).

Simplification reduces complexity to focus and attune attention to particular features of a social process, while necessarily de-prioritizing other elements (Healy 2018; Leifer 1992). This process often involves decisions about classificatory boundaries and labeling those distinctions (Wickham et al. 2023; Healy and Moody 2014).

Researchers' *theoretical interpretations* bring empirical observations and broader theoretical frameworks to bear on one another, establishing "what is this a case of?" (Ragin and Becker 1992). In this use, visualizations help identify what we have learned through the lens of theoretical expectations, including what our data reinforce, fail to support, or suggest requires further refinement (Borgatti and Halgin 2011; Liu et al. 2017).

Scientific principles for visualization practice

Before turning to how visualizations can optimize these purposes across particular tasks, I describe two additional scientific principles that govern how researchers pursue these ends, which should guide any (network) data visualization.

Network visualizations often present unique *ethical* challenges, particularly concerning potential *confidentiality* and *deductive disclosure* (adams 2019; Ortiz Ruiz et al. 2026). The potential for deductive disclosure is broad in any presentation of multiple variables (Rocher et al. 2019), especially in the "closed population" settings common in network research (but see also Zimmer 2010). Relational data provide additional avenues for serial deductive disclosure (Narayanan and Shmatikov 2009).[6] To this end, the frequent practice of plotting empirical networks from data collection efforts may undermine principles of confidentiality and minimal risk, which should lead us to consider alternative visualization strategies that can leverage visualization's strengths while minimizing those potential risks (Ortiz Ruiz et al. 2026). These limitations do not preclude network researchers from visualizing, and even sharing, data (Pasquale et al. 2025). Rather, they suggest that we

[6] I.e., the revealed identity of one node in a graph (e.g., myself) can in turn heighten the capacity to identify others (e.g., my friends).

may need to tailor our approaches in ways that current practices do not necessarily optimize.

While the benefits of open-science principles are not new (Partha and David 1994), researchers have recently (re-)discovered enthusiasm for better implementation in practice (Miguel et al. 2019).[7] *Replicability* therefore raises particularly thorny questions for (network) visualizations.[8] Network visualizations entail a combination of node/tie inclusion and exclusion rules, their respective attribute information, and determinations of the positioning of each. This forces consideration of what scope of information (i.e., which variables for which nodes and which ties) and granularity of presentation (e.g., levels of measurement per observation) we include in visualizations.[9] In particular, visualization's utility may not require including the full complement of information available in the data. Additionally, because positions in network visualizations are typically defined relatively (vs. absolutely), algorithmic and procedural details (e.g., random-number-generator seeds) can be important for faithfully recreating visualizations that are substantively and/or aesthetically comparable (Bender-deMoll and McFarland 2005).

Network visualization across research stages

While research processes entail many potential stages, three commonly employ visualizations as a primary tool. **Table 1** outlines a heuristic typology of visualization strategies that roughly correspond with different stages of research, and consider how they each differentially prioritize the purposes and accommodate the principles described above. To foreshadow, this differentiation leads to a suggested strategy of visualizing *modeled analytic results* as a unique combination that remains under-used in network research—a practice I outline in the following section.

Table 1: Potential Comparative Benefits across Visualization Strategies

purpose or principle	observed	conceptual	modeled
describe / explore	✓	?	x
simplification	?	✓	✓
case interpretation	✓	x	✓
theoretical interpretation	x	✓	✓
confidentiality and/or deductive disclosure	x	✓	✓
replicability	?	✓	✓

✓ = generally advantageous; ? = context-dependent; x = generally ill-suited.

Much of the data visualization literature assumes *sequential* stages, with a corresponding set of visualization

[7] In fact, in personal communication, one colleague noted that the "sociogram is ... one of the purest original forms of open science, [... and] replicable data transparency."

[8] While the existing literature sometimes uses "replicable" and "reproducible" interchangeably, and others distinguishing them, those distinctions have at times been employed in exactly opposite ways (Miguel et al. 2019). For clarity, I use only *replicability* to indicate the ability to recreate a visualization in the same way as the original, whether data-linked or otherwise.

[9] Given the aims of this paper, I focus on the replication of *network visualizations* independently of the numerous additional appropriate considerations for how to implement replicable and reproducible research more broadly (Miguel et al. 2019), including the particular demands of network research (Neal et al. 2024; Pasquale et al. 2025).

strategies asserted to align across them. To avoid wading into epistemological and theoretical debates about research processes,[10]–here I focus on the potential orientations to visualization, and simply note the stages of research in which these are often suggested by data visualization scholars.

Observed network visualizations provide a means for presenting the case(s) at hand. This approach typically prioritizes fidelity to the features in the data, accounting for what they contain (Bender-deMoll and McFarland 2005). Such descriptive approaches are especially common in explanatory data analysis (EDA) stages (Wickham et al. 2023; Healy 2018). While EDA be framed as a "first step" toward examining and presenting one's case, it does not mean that the visualization directly "reflects" reality. It merely indicates that the data have been obtained, minimally processed, shaped into a coherent presentation strategy, and presented to the reader (Wickham et al. 2023; Healy 2018).

Visualizing observed data provides opportunities to develop familiarity with the details of data–whether as researchers make sense of what they will be analyzing (Martin 2018), or in presentations of those features to readers. The fundamental steps to generating observed network visualizations are: (1) obtaining the data (through primary methods or secondary sources), (2) pre-processing, cleaning, and organizing the data, (3) reducing to focal elements (relational and nodal attributes), (4) transforming that reduction to visual form, and (5) labeling it. Many of the pre-processing, presentation, and sense-making steps involved at this stage benefit from intimate knowledge of the data (Potterat et al. 2004). Although this knowledge can inform researchers' work, it may undermine principles of confidentiality and/or deductive disclosure in ways that may not be advisable or permitted in publishable or directly replicable forms (Ortiz Ruiz et al. 2026).

For networks, visualizing observed data can characterize patterns of general node, tie, and/or network patterns of composition and structure and by identifying potential needs for analysis (Freeman 2000).[11] A common use of these visualizations is to serve similar purposes to the "descriptive statistics" Table 1 that many papers provide—giving the reader a means to orient to the case at hand–both providing overview summaries, and indicating details that help make sense of the particularities of the case(s). However, publishing observed network visualizations can easily lead to visuals that are overly complex in ways that obfuscate what we can learn from them, draw attention to the particularities of the case, rather than the general principles the case is examined to characterize, be difficult to precisely replicate, raising questions about consistency in interpretation, and introduce potential risks that could violate research ethical standards. When used for exploratory purposes, the necessary simplifications of visualizations can sequentially emphasize

[10] e.g., I'm allowing for applications of the approach proposed here in inductive, deductive, *or* abductive orientations Timmermans and Tavory (2022).

[11] For example, this entails the pre-processing steps of variable reclassification or anomalous-case handling, (Wickham et al. 2023).

different dimensions across multiple iterations, while "rotating" the priorities across them to explore multiple simplifications of the data's high dimensionality. Although primarily descriptive, using observed network visualizations as exploratory steps can also bridge to the analytic aims of research, seeding potential analyses for more inductively oriented work or beginning the process of case interpretation in more deductively oriented work (Wickham et al. 2023).

Some fields widely incorporate visualizing **conceptual** frameworks, including those with varying degrees of formalization—ranging from those that primarily serve to "organize one's thoughts" (Brady et al. 2020) to approaches that formally articulate modeling aims and assumptions [e.g., directed acyclic graphs; Elwert (2013)]. Network research has periodically leveraged such visualization aims to specify how *combinations* of features can lead expectations to diverge from various "ideal type" formations of networks. Visualizations can make these configurations concrete in ways that may not be as readily intelligible without them. Small world literature provides a compelling case of this form - plotting "cave man" graphs with only a few cross-group "shortcuts" necessary to shorten average path length (traditionally associated more with random networks), while maintaining high overall clustering (Watts and Strogatz 1998).

Conceptual visualizations generally characterize the research's selected modeling aims (adams 2020), and entail steps that (1) articulate the research's theoretical claims, (2) which elements are not focal, but are included in the analytic approach, (3) what assumptions remain in the analyses (and any attempts to assess how robust estimates would be to those assumptions), and potentially, (4) things that are explicitly asserted to not be implicated in the examined questions. This can help to formalize which theoretical claims a study would have the ability to support, disconfirm, or suggest alternate interpretations.[12]

In their "pure" form, conceptual visualizations tend to prioritize theoretical interpretation and simplification over all else. Additionally, given the potential for (nearly) "data-free" visualizations of this sort, ethical considerations are likely minimal.

Visualizing **modeled** results[13] allows researchers to highlight the research's *theoretical* implications (Rawlings et al. 2023). For example, in social network research with explanatory (e.g., causal) aims, researchers seek to demonstrate how observations in empirical cases (Kadushin 2012) combined with appropriate analytic strategies (Borgatti et al. 2024; McCarty et al. 2019), conform to expectations in ways that support theoretically appropriate interpretations (Borgatti and Lopez-Kidwell 2011).

Because an emphasis on theoretical representation often highlights general patterns or "conceptual gener-

[12]Here, visualizations are performing the same clarifying and formalizing task that is accomplished when we employ "writing as thinking" (Zinsser 1993).

[13]As above, "modeled" is not limited to statistical approaches.

alizability" (Collins et al. 2024), this approach need not rely on the specifics of the data–thus allowing researchers to readily maintain confidentiality and facilitate replication (Bender-deMoll and McFarland 2005). Alternatively, researchers may primarily draw on theory and analysis to account for events in particular cases *per se*. Even in this case, simplifications are still necessary over "full" data fidelity, identifying certain focal aspects to make sense of the case while de-emphasizing others. This also points to a further caveat–while many data visualization primers assume theoretical interpretations necessarily target causal (explanatory) goals, data visualizations can represent a wide range of analytic results that may instead prioritize understanding (*verstehen*) or descriptive aims.

In sum, different research stages and visualization strategies incorporate different optimizations across the purposes and principles of network visualization. Given the potential for distinct combinations across these multidimensional aims, I next summarize how their intersections suggest alternative orientations to visualization strategies, draw a few parallels to existing practices, then outline the steps for implementing modeled network visualizations.

Implications for visualization practice

This potential for non-overlapping combinations can produce sub-optimal visualizations in several distinct ways. Perhaps most obviously, when researchers construct a visualization to serve one of these purposes (e.g., exploratory data analysis), it will not optimally serve other aims (e.g., explanatory) because the combinations necessary to achieve one task well can trade off against the ability to optimize for other purposes.

Perhaps more problematically, when researchers *assume* (or worse, *require*) that visualizations have "universal" benefits, they presuppose that the purposes of data visualization are necessarily shared across this range of uses. One common version of this approach in SNA produces a "descriptive" plot of the observed data and then "overlays" features of the theoretical interpretation of analytic results, in hopes that readers will recognize the "important" signal amid the remaining noise in the represented data. Unfortunately, this type of compromise strategy can fail to accomplish *any* of our aims well (Pache and Santos 2010). My contention is that network visualizations not infrequently commit one of these two errors.[14] This leads me to propose an alternative visualization strategy that optimizes on a combination of purposes not adequately served by our most common approaches, while maintaining the visualization principles detailed above. In particular, I propose that "modeled network visualizations" can more readily combine theoretical interpretation with empirically grounded simplification while preserving confidentiality and facilitating replicability.

[14]Self-incriminating example removed for review.

Modeled network visualizations

My fundamental proposition is simple: rather than relying solely on "observed" networks, we should develop a strategy for developing "modeled network visualizations" - especially for presenting analytic results. This is not a ground-breaking suggestion. In fact, many statistical modelers—of social network and other types of data—already enact this practice, but publications do not consistently deploy this approach in ways that fully leverages its potential utility. For example, "eye-balling" consistency of modeled results against observed network visuals—in tandem with formal statistical tests for goodness of fit—is typically taught as a helpful practice in model development and fitting processes (Krivitsky et al. 2023).[15] However, while we use these heuristic network visualizations to improve model fit or in other analytic development steps, authors still often resort to presenting the observed version as their primary network visualization(s) in publications. I have done this multiple times myself.

"Modeled network visualizations" can provide a better alternative when we seek to characterize analytic results in ways that better achieve certain combinations of the aims and principles outlined above.[16] In describing the proposed approach and examples below, I focus on the conditions under which modeled network visualizations offer relative advantages in pursuing this unique combination.

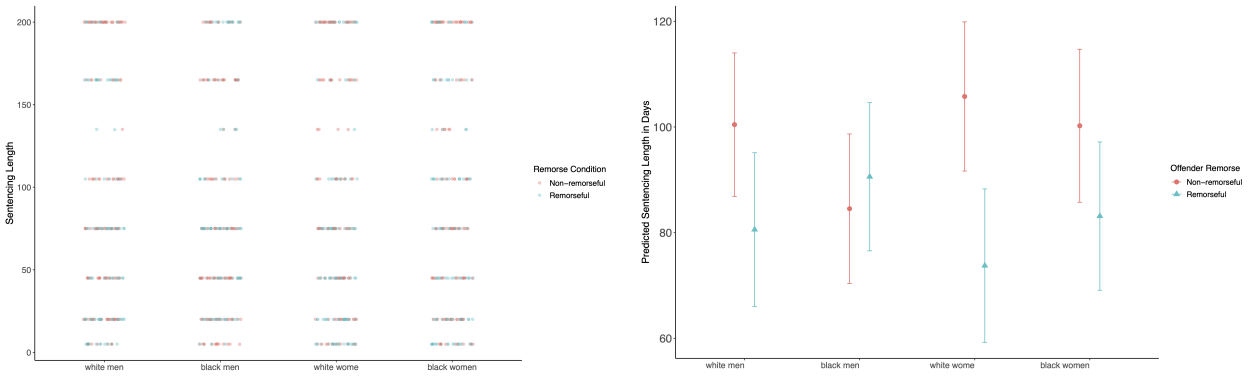
Effect plots - a parallel motivation

Let me draw a few parallels to the approach I propose from strategies researchers frequently employ that leverage (parts of) the same logic–especially in presenting and interpreting modeled results. Logistic-regression coefficients (as with ERGMs or SAOMs) are notoriously hard to make substantive sense of on their own (e.g., significance and magnitude of effects are frequently conflated in results interpretations by writers and readers alike). Interaction terms in these–and other forms of–regressions are similarly difficult to interpret. A common solution has been for researchers to: (1) gather, pre-process, and provide a descriptive summary of their data to give an overview of what the sample being analyzed represents and how key focal features are distributed in the population; (2) conduct statistical analyses to investigate the researchers' theoretical aims, which are summarized with traditional combinations of betas, standard errors, log-likelihoods, etc. to represent the fit of those claims to empirical estimation; and then (3) transform those modeled results into "predicted probabilities" in ways that help interpret what those modeled results "look like" in their implications (Fox and Andersen 2006).

To "tell the story" that arises from the model results, we do not return to the raw data, provide the

[15] They can be especially helpful in identifying theoretical limitations of existing models or enhancing stability of estimates.

[16] I do not also contrast these with "conceptual" visualizations in what follows, because researchers often develop those intentionally in *non-empirical* ways.



(a) Descriptive Plot, Outcome x Conditions

(b) Marginal Effects Plot, with Interaction

Figure 1: Predicting Sentencing Length from Expressed Remorse, split by Offender Race & Gender (Zhao & Rogalin 2024).

individual-level combination of variables for some subset of the population, and rely on those cases' patterns to convey the implications of our research (in the rare cases that people do, they are often "synthetic" cases, not real ones, e.g., Ayala and Koch 2019). In **Figure 1**, the right panel makes apparent that "remorse" benefit in sentencing reductions applies across groups, except for black men (Zhao and Rogalin 2024). Deciphering that result from the left panel would be next to impossible; but we also would rarely attempt to use a descriptive plot for those purposes. To do so would not only fail to clearly tell the aggregate story we have learned in the course of theorization and analysis.[17] So why, in network research, do we so often feel uniquely compelled to present the "real" pattern of relationships from our data as a means to interpret and convey our findings in visual form? The approach I describe below generally imports these existing practices into network visualization practices.

Blockmodels, motifs, & coefficients - partial *network* parallels

I also note that the basis of the general idea—-if not the complete practice–I am recommending, is already in use in some domains of social network research. Perhaps the most direct corollary is how blockmodeling results tend to be presented as a representation of role profiles (Doreian et al. 2005). The image-matrix and/or reduced-graph presentation distills the overall role structure of an analyzed graph into summary form (Borgatti et al. 2024). A permuted adjacency matrix arranged by density blocks maps that summary onto the full data, but typically does so for the *observed* adjacency matrix rather than for a modeled representation. Other "coarse-graining" visualization strategies for "large" networks commonly take similar parallel approaches (Newman and Girvan 2004).

[17]It is additionally well known that overly granular descriptive statistics can inadvertently pose deductive disclosure risks, thus leading to "cell size" thresholds for many reporting standards (Rocher et al. 2019).

Another approach to leveraging visualizations to characterize analytic results in theoretically -focused ways exists in *network motifs* (Schwarze and Porter 2021). Network motifs identify *repeated* structural patterns across multiple (often, but not exclusively, *personal*) sub-graphs of networks. They analytically identify these patterns as abstracted "common forms," that are often presented in distilled and summarized form, rather than presenting multiple parallel observed networks for the full sample, and highlighting overlaid similarities across the repeated panels. These aims are similar to those in blockmodels, though network motifs typically consider *local* configurations. However, network motifs—and their visual style of presentation—remain more widely adopted in network science than among social network researchers to date (for an exception see Vacca 2020). Other approaches in ego-network research similarly draw on stylized representations to visualize analytic implications, rather than reverting to empirical networks to explain findings (especially when doing so would violate confidentiality, McCarty et al. 2019).

Such configurational representations have also been employed to assist with interpreting the results of inferential statistical models of social network data by presenting visualizations of micro-patterns rather than coefficient names in results tables (e.g., in stochastic actor-based models, Snijders and Steglich 2026). The approach I describe below generally expands these applications to apply to visualizing networks as wholes from these applications to visualizing elements of social network patterns.

Implementing modeled network visualizations

Although modeled network visualizations can take many forms,[18] their implementation follows a similar structure across these potential forms. This section elaborates the four common stages of this process. It is important to note that of the stages I describe here, only the fourth is explicitly about the *visualization* strategy I propose here; steps 1-3 are the analytic steps that are assumed to have been completed *before* visualizing.[19] I accompany the description with an illustrative example drawn from a strategy for teaching and model building in exponential random graph models (Krivitsky et al. 2026; Lind 2012). I then illustrate the application of this approach through an extended empirical example.

Step 1: Characterize the data to be examined

As with the recent resurgence in attention to the importance of *descriptive* statistics before conducting and presenting any multi-variable analyses (Ayala and Koch 2019; Healy and Moody 2014; Wickham et al. 2023), the first *empirical* step in research is to generate a preliminary account of the data to be examined.

[18] Recall that I use "modeled" in a theoretical and conceptually based sense, not necessarily a statistical one.

[19] As such, the paper is not concerned with identifying the appropriate analytic approaches for one's research questions, nor diagnosing whether those have been appropriately conducted. That is a separate step of the research process that is beyond the scope of this paper, and is better addressed theoretically and analytically.

The full details of what this should entail likely depend on the scientific aims and nature of the data to be investigated (e.g., qualitative or quantitative, closed or open population, etc.), but scholars have recently suggested potentially standardizing these to include a variety of features at the node, tie, and network levels (Neal et al. 2024). In some—but, as with my proposal here, *not all*—cases, an observed (network) visualization can enhance this descriptive task. That is, my proposal doesn't exclude exploratory network visualizations, instead asking us to reconsider if they're right for *other* (or all) purposes.

The observed network in panel A of **Figure 2** provides a visualization of the adapted pedagogical example I draw on below while developing this strategy. This example presents the "hookup" network among characters in *Grey's Anatomy*, which I will estimated using an exponential random graph model (ERGM) and visualize its results. This example is mildly adapted from a widely used example developed by Benjamin Lind (2012).

Step 2: Conduct analysis appropriate to the research question

The second step is to conduct whatever analyses are appropriate to the theoretical aims of the research questions and the nature of the analytic dataset. This step reiterates a fundamental feature of the proposal here: "modeled" results come in many forms. The *Grey's Anatomy* example draws from *statistical* models (ERGMs, in particular), but the approach applies equally to *any* other form of data-analytic results—ranging from similarly statistically oriented models such as SAOMs through to the primary takeaways from grounded-theoretically oriented ethnographic observations (Dominguez and Hollstein 2014).

Building (statistical) models of observed network data can be a complicated process. In developing "well-behaving" models, estimating of goodness of fit can assist with interpreting what the latest model estimates are—and are not—accounting for in the data. Many tutorials for developing such models employ visualizations of a single network instance from intermediate model results, both as a means to intuit how models work when teaching–a necessary step when enhancing model estimation and fit in real analyses (Krivitsky et al. 2026).

Examining the observed network above in light of theoretical goals,[20] suggests a variety of model parameters to investigate. In the *Grey's Anatomy* example, we could fit models that include degree effects; race, gender, or age effects (both as "activity" and "homophily" parameters); or structural features (e.g., local closure rules (Lind 2012; Krivitsky et al. 2026)).[21]

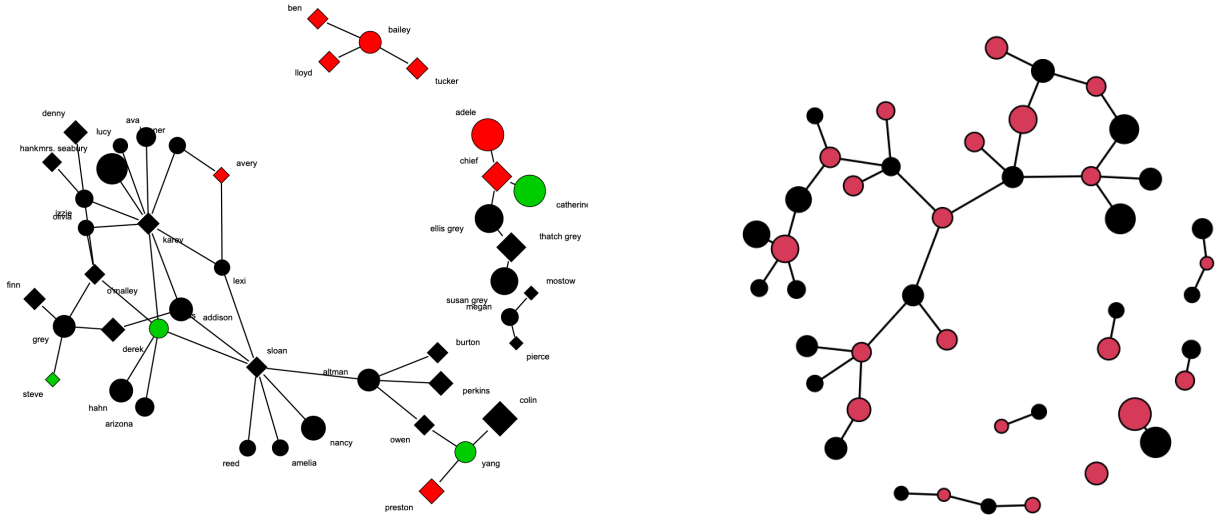
[20] In empirical work, this combination likely drives both the data we collect in the first place and the elements we draw from it in exploratory/observed network visualizations.

[21] In **Figure 2** Panel A, "isolates" are not present because they were excluded by definition in the way the data were encoded, so a restriction against isolation should be part of the model.

Step 3: Compile an appropriate summary of analytic results

Analytic results and interpretations are necessarily reductions of the full details available in our data, and do so for differentiated purposes (adams 2020). Done well, they draw our attention to the scientifically useful features of the data, lead us to appropriate interpretations, and limit the distractions that keep us from achieving those aims—regardless of the particularities of the case at hand (Healy 2017). The third step is therefore to compile an appropriate overview of analytic results in summary form (whether statistical or otherwise).

Drawing on the *Grey's Anatomy* example, the fitted ERGM supplies this summary, including both model-specification details and corresponding results for the network features it identifies as important.[22] The results elements of the *Grey's* example include gender *heter*ophily, age similarity, and a restriction on isolates.[23]



(a) observed network **(b)** modeled network

Figure 2: Two visualizations of *Grey's Anatomy* hookup network. Panel A represents the "observed" data with identities and multiple attributes attached (size=age, color=race, shape=gender). Panel B represents a network generated from the modeled results, including only those exogenous features that are part of the model (node color=gender, node size = age). This example is adapted from Lind's (2012) ERGM tutorial.

Step 4: Generate a modeled network from that summary

The fourth step is where my proposal comes in–with results in hand, we then convert that summary into a "predicted" network that conforms to the key features of the estimated analytic results. This modeled

[22] These should include *all* details necessary to estimate *and* interpret the model. In the case of ERGM, these are returned as part of the model object and include such information as model coefficients and standard errors, goodness-of-fit statistics, and convergence diagnostics (Krivitsky et al. 2023).

[23] This excludes also investigated node-effects for gender, race, or age, and a racial homophily term. Since formal goodness-of-fit statistics indicated including these produced more poorly performing models, they are excluded from the final results.

network visualization will necessarily differ from any observed network visualization by eliminating features that are not "modeled" (Schwabish 2021), reducing the prominence of idiosyncratic variation (i.e., identifying "what is this a case of?" (Ragin and Becker 1992; Collins et al. 2024)), and highlighting only the key elements of the findings that warrant theoretical interpretation (Vertesi 2025).

Such modeled visualizations therefore provide: an empirically anchored, theoretically prioritized, network visualization that clearly communicates the researchers' key interpretative takeaways–and ideally in ways that optimize what visualizations convey that statistics alone could not.

In **Figure 2**, panel B visualizes a single draw from the ERGM results for the *Grey's Anatomy* analysis, allowing comparison with the observed network in panel A. The modeled visualization includes only the aspects incorporated in the model, which in turn diminishes the representation of idiosyncratic aspects in the data.[24]

Panel B of **Figure 2** also illustrates a number of features that are instrumental to the model-fitting process from which this example derives, but nonetheless differentiate modeled network visualizations from other strategies. In terms of explanatory capacity, names do not diminish the ability to observe important features of the network (e.g., a general, but not absolute, heterosexual partnering pattern). Noe that the nodes in panel B are note merely "label removed" but by deriving from a modeled network, the nodes are "characteristic" and there for do not provide one-to-one mappings to real data points. This abstraction helps draw our attention to the interpretive implications of our analysis rather than toward the idiosyncrasies of the case at hand (Becker 2017). This may raise the question of whether visualizing a single draw from modeled results is idiosyncratic in some (non-empirically derived ways). This concern belies the parallel realization that observed data also provide only a single, characterization of reality, invariably including error along with the information they convey.[25] Our confidence in the accuracy of *both* observed data and modeled results are that the key *theorized* and empirically-tested features they embed are robust to any such errors (Becker 2017).

Beyond this theoretical simplifying focus, the *Grey's Anatomy* example also illustrates the confidentiality preserving nature of the modeled visualization approach. If these were actual data carrying confidentiality concerns, the modeled visualization—and its rendering of modeled results as unlabeled, "representative" nodes and ties rather than a capture of actual relationships—would also remove risks of sequential deductive disclosure.[26] Finally, because the visual is based on model results, the model-results object is likely shareable

[24]Here, the visual specifically excludes node labels and the race nodal attribute because goodness-of-fit statistics confirmed that these did not contribute to the observed relational pattern. More formal goodness-of-fit tests also confirm the "visual" similarity between the general structures of the two graphs, while other, lesser-performing models showed obvious divergences, visually *and* statistically (Lind 2012).

[25]Treating the realized particulars of that instance as central can over-reify what is often only an approximation–though ideally a reasonable one–of the theoretical question being investigated (Becker 2017).

[26]For example, given the infrequency of same-sex or inter-racial relationships in the population, even an *unlabeled* version of

even if the raw data are not,[27] facilitating the replicability of both the results and the visualization.

An illustrative empirical example - Rap Samples

Finally, I demonstrate how the visualization strategy I propose could be applied to an empirical analysis using a quite different analytic approach from the ERGM example above. In addition to showing how a modeled visualization can help facilitate interpretation of research takeaways,[28] stepping through the process for this case allows me to articulate how future research can hone this practice. First, rather than providing a "recipe" to apply uniformly across all use cases, this approach structures a set of *principles* that can be applied to a range of particular data formats and analytic strategies. Second, at various stages in developing the example, I will highlight *sensitizing considerations* regarding what this approach is optimized towards, and some of the limitations it contains.

Jennifer Lena mapped rap songs' sampling of other artists (Lena 2004). She found (1) a small "core" set of sampled artists from whom a disproportionate volume of samples were drawn, and (2) "circles" among sampling artists that: (a) differentiated among sampled artists by source genre, and (b) distinguished groups of sampling rappers through their sampling repertoires, with those groups also partially sorted by rap sub-genre. Later work also showed that these patterns were partially shaped by industry differences (Lena 2006) and contributed to the development of status hierarchies among the sub-genres (Lena and Pachucki 2013).

Following the steps outlined above, we can produce a corollary modeled network visualization, despite these analyses not relying on a single "formal" statistical network model. The papers each provide a number of descriptive features in the data (step 1 above), including the frequencies, temporal trends, and "key players" (song and artist level). Panel A in **Figure 3** provides an exploratory bipartite plot, with sampling songs represented by triangles and sampled artists by circles. This plot begins to characterize some of the features of these sampling data–especially noting the differential rate at which sampled artists are sampled (node size of circles), and the concentration of that activity among select genres (e.g., the larger nodes are disproportionately green). Additionally, this plot is suggestive that the network is characterized by a core-periphery structure. Other elements are much less visually apparent without first conducting, then representing systematic analyses–including whether it is governed by community structure and/or genre sorting.[29]

the graph on the left would allow partners of Derek or "Chief" to likely identify *themselves* in the plot, and thus "discover" their partners having other partners.

[27] It is not uncommon in social network projects for participation agreements to have data-sharing restrictions on the *relational* aspects of the data, as a means to ensure confidentiality.

[28] I note that neither of the primary empirical papers on which this example draws includes network visualizations. In part this reflects the complicated multi-tiered relational patterns described win those analyses. As such, this use case is primarily illustrative.

[29] Some authors return to exploratory visualizations such as this one and layer analytic properties onto them. I provide a

Lena's conclusions arose from a sequence of descriptive and interpretive social network and non-network analyses (step 2 above), yielding empirically supported theoretical conclusions that can be summarized (step 3). Focusing only on the network elements of those results, these describe networks characterized by: (a) highly skewed degree distributions among sampled artists, and (b) a strong core-periphery structure, with a relatively small set of sampled artists forming the core. They also suggest community structuring at two levels: (c) among sampled artists, with communities partially sorted by source genre; and (d) among sampling artists, with communities partially differentiated by the source artists and songs in their sampling repertoires, which are partially sorted by rap sub-genres.

Let me reiterate a note from above, up to this point–i.e., through the first three outlined steps –there is nothing new in the process beyond typical research processes, producing summarized analytic results. Next, I turn to how these summarized network properties can be translated into a network visualization characterizing these model results (step 4).

Translating these findings into visualization requires formalizing any modeled properties into visual components.[30] Panel B of **Figure 3** presents the modeled visualization addressing the combination of results from Lena's work as summarized above. I leverage this visualization to elaborate the strategy's relevant principles and sensitizing considerations, following the steps common to all network visualizations: defining the node set(s), edge set(s), layout, and visual layers (e.g., color, shape, and size).

Step 4.1 Defining the node and edge set(s)

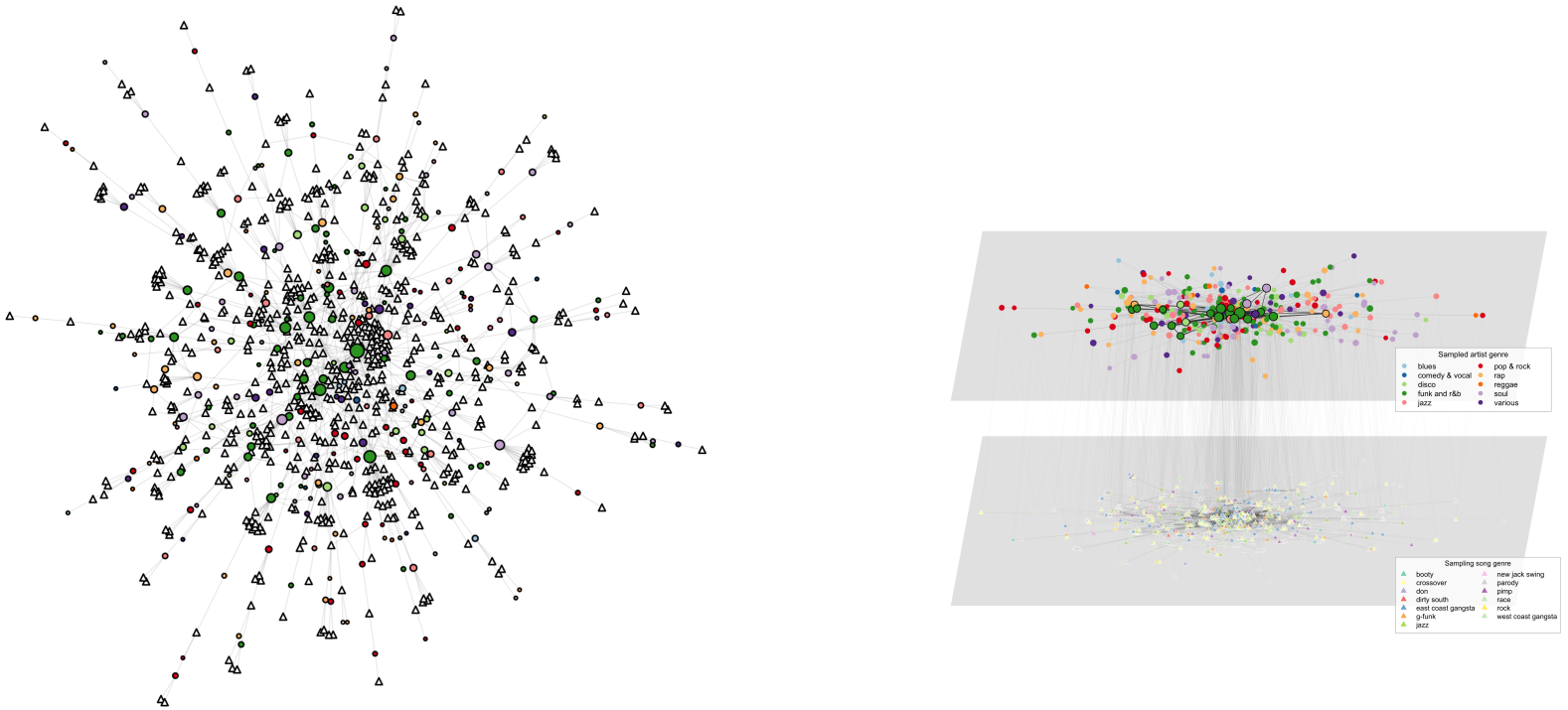
Lena's data contain 1604 samples linking 964 rap songs to 492 unique sampled artists. I therefore begin the modeled visualization by matching the sizes of these two node sets.

I retain Lena's bipartite data structure, in which ties connect sampling songs to sampled artists.[31]. The visualization also incorporates two analytically constructed within-mode relational counts: overlap among sampled artists within songs and overlap among sampling songs through their shared sampled artists, built from standard two-mode to one-mode projections. An observed network visualization would therefore represent the original cross-mode sampling edges and the corresponding projection counts: (a) how many times the same artists are sampled within the same song and (b) how many artists are jointly sampled between pairs of rap songs. For generalizing the approach, note that this set *includes* one edge type from the original data, *constructs* two edge types, and *removes* one element from the original data (while sampling

corresponding version of this approach in the appendix as a hybrid of the exploratory and modeled strategies presented here.

[30]In the ergm example above, this builds networks that have statistics corresponding to each of the included statistical coefficients.

[31]An alternative would be to convert these data into a single node set with node type encoded as a nodal attribute



(a) exploratory visulization

(b) explanatory visualization

Figure 3: Two visualizations of Jennifer Lena's (2004) rap sampling network. Panel A represents the "observed" data, including sampling/sampled artists (shape), sampled degree (size), and genre (color), with a few artists in prominent positions labeled. Panel B represents a network generated from the analytic results, with the sampled degree, artist genre, and multi-level community structure highlighted.

edges were weighted–some songs sampled the same artist multiple times–those weights are not included analytically or visually).

Thus, the modeled visualization requires constructing three distinct edge sets. First, within each mode, the nodes are assigned a sampling strength, according to the distribution for each node class from the observed data. In the sampling song projection, those are randomly assigned; in the sampled artist projection, they are assigned according to the genre-specific distributions from the empirical data. Together, these result in edge "stubs" that are not yet matched (Frank 1981)–i.e., are not yet created *as* edges (Hunter et al. 2008).

Assigning these stubs as edges is then an additional two-stage process. The first stage draws on empirical degree distributions for the cross-mode sampling ties and empirical strength distributions for the within-mode projected ties. The second allocates ties through mixing patterns that variously combine strength assortativity and genre mixing. For the original sampling edges (across mode), I use the empirical "mixing matrix" between sampling rap song genre and sampled artist genre, which is degree-conditioned–*only* on the sampled artist mode. Within the sampled artist projection, stubs combine the empirical strength-assortativity associations (reflecting Lena's findings of a core-periphery structure), with the empirical genre mixing matrix (noting that the likelihood of being jointly sampled is not only a factor of genre, but also of sampled artist popularity). Finally, within the sampling-song projection, ties are allocated to correspond with the empirical genre-based mixing matrix, without conditioning that mixing allocation on sampling-song strength.

This illustrates a process for mapping the analytic results onto the visualization, while also excluding information that was computed, but not relevant to the findings. Notably, these allocations match several of the key analytic features from Lena's analysis, including the genre-based partial structuring of sampling, both across node classes (the samples) and within mode (patterned sampling pattern overlaps).

Step 4.2 Selecting a modeled realization and layout

Having specified a family of networks consistent with the modeled properties as generated in step 1, the next step is to select one realization from that family and produce a layout for it. All network visualizations—empirical, modeled, or otherwise—are individual renderings selected from a wider set of potential layouts. Moreover, those visualization instance(s) each carry error from a variety of potential sources, including measurement error in empirical networks, stochastic noise in layout algorithm's iterated optimizations, or model-based error (e.g., in the form of estimates' standard errors). Accordingly, any visualization strategy must explicitly select its layout strategy to align with the nature of the data, and reflect the core features intended to be highlighted–i.e., modeled network visuals should optimize conveying the analytic summaries produced in step 3 above. Additionally, simplifications should reduce or remove any idiosyncratic elements

from the visualization to reduce their detracting from observing the visualizations' key implications–e.g., not including unexamined details in observed networks, or un-analyzed or unimportant (e.g., non-significant) elements in modeled visualizations.

For the results described above, this translates into two key layout choices. First, I present the modes as separate network layers. This allows the layout to simultaneously capture elements of group structuring - those operating among genres within layers (modes), and the partial structuring between genres across layers (modes). Relying on a bipartite graph layout would necessarily reduce these three dimensions of genre structuring–within the sampled-artist layer, within the sampling-song layer, and across layers–to two, preventing the visualization from conveying Lena's multidimensional genre structuring. Second is the choice of layout algorithms. Here, I used a multi-layer approach that allowed for treating within and across layer ties with different weightings, to preserve the clustering patterns that happen separately within and across layers, but not treating them independently. In addition I chose a layout algorithm that optimizes local densities rather than global network distances to prioritize this group (genre) structuring. Given the local optimization involved in visualizing complex networks, I generated multiple realizations and layouts, selecting one that preserved the relevant core-periphery and genre-clustering patterns while neither overly determining those patterns (e.g., by exaggerating the separation between genres) nor introducing unnecessary idiosyncratic noise (e.g., layout positioning) that detracted from conveying them.

As can be seen in Panel B of **Figure 3**, this layout makes two features of the overall network structure legible. Most prominently, each layer exhibits a core-periphery structure; in the sampled-artist layer, that core is composed of highly sampled artists. Additionally, both within and across layers, there is minor additional structuring by genres–in the sampled artists layer (top) see that pop-rock (red) shows a slight concentration on the south-west portion of the graph and the jazz (pink) to the east, while in the sampling song layer (bottom) the east-coast gangsta genre is located predominantly on the eastern portion of the graph. Also note the "streams" of sampling ties flowing between layers suggesting a further cross-layer manifestation of the same partial genre structuring as Lena identified (contrasting with the possibility of more random/criss-crossing arrangements across layers).[32]

[32]In supplementary visuals in the appendices, I also provide 3 additional views of these layouts to further illuminate these patterns–one each allowing the respective layer's layouts to optimize individually, and a third that is a "top-down" rotation of the included figure using an edge-based density contour approach to better highlight the separable streams of ties flowing between the mode layers.

Step 4.3 Layering attributes for details and emphasis.

The final step is to layer analytic information onto the modeled network and its layout. I represent sampling degree in the sampled-artist layer proportional to node size and genre in each layer with node color.[33] Additionally, ties among sampled artists have thickness proportional to their sampling overlap. Finally, to avoid distracting from the key takeaways, individual inter-layer edges are substantially de-emphasized; their aggregate streams nevertheless remain visible.

Interpreting the visualization's scope

To recap, the resulting modeled visualization encodes each key finding summarized in Step 3: the skewed degree distribution among sampled artists (node size); core-periphery structure (radial pattern); strong overlap among members of the sampled core (edge thickness); and partial genre sorting across three dimensions (minor spatial clustering by color within layers, and strong patterning across layers). While the visual still contains uncertainty (e.g., that the genre sorting appears relatively weak, but present), and some idiosyncrasies (e.g., stochastic variability), this differs from things that are fully absent from the visualization.

The potential bases of such exclusions are worth considering. In many cases, analytic results will conclude certain features are not important to the process being investigated (e.g., statistical non-significance), and thus should not be visualized. There are other features that may be assumed, unavailable, or uninvestigated; as with omitted-variable bias in analytic results, such features may nevertheless structure the patterns shown in a visualization in ways that cannot be derived from the modeled results.

I also want to highlight an advantage of the approach that could be readily visible here. While node-labels may assist in case interpretation for an observed network visualization–e.g., in this case we could have highlighted the largest nodes in the sampled layer being James Brown and George Clinton—but those details are not necessary to convey the implications of the modeled results. Coreness, the skewed distribution of sampling, relationships among core artists, and differential genre allocation can all be represented by a synthetic set of artists without relying on their identities. Although this particular example presents no disclosure concerns,[34] it illustrates the broader possibility of communicating theoretical implications without labels in other cases where disclosure risks would be present.

[33] Lena encoded separate genre sets for sampled artists and rap sub-genre for sampling songs; I apply distinct color palettes to the two layers.

[34] I selected this example in part explicitly because it allows comparison of labeled and unlabeled visualizations.

Quo Vadimus?

As with any visualization strategy, network visualizations can be implemented well or not so much. Many familiar failures—unintelligible "hairballs," poor layout choices, over-complicated visuals, and other visualization "sins"—stem from poor implementation *per se* (Tufte 1983). But a different weakness arises when we treat visualization as a single task that we assume can be optimized by global principles. A visual strategy that serves one purpose well may be poorly suited to another.

I agree that visualizations are rightly one of our field's distinctive calling cards (Freeman 2004), reflecting their utility as part of the social network research toolkit (Freeman 2000). However, I also contend that their ubiquity has increasingly led us to occasionally use them as the proverbial hammer, even when the approach we adopt may not be well-suited to the task at hand. As such, I'm provoking the field to consider whether modeled network visualizations are worth adding to our toolkit.

I'm not seeking to replace exploratory or case-based visualizations. Rather, for particular combinations of scientific purposes and principles, modeled network visualizations can more effectively convey the theoretical implications of analytic results, while enhancing their capacity for replicability and confidentiality.

References

adams, jimi. 2019. *Gathering Social Network Data*. Quantitative Applications in the Social Sciences 180. SAGE.

adams, jimi. 2020. "What Are COVID-19 Models Modeling?" *The Society Pages*.

Ayala, Ricardo A., and Tomas F. Koch. 2019. "The Image of Ethnography—Making Sense of the Social Through Images: A Structured Method." *International Journal of Qualitative Methods* 18 (January): 1609406919843014. https://doi.org/10.1177/1609406919843014.

Becker, Howard S. 2017. *Evidence*. University of Chicago Press.

Bender-deMoll, Skye, and Daniel A McFarland. 2005. "The Art and Science of Dynamic Network Visualization." *Journal of Social Structure* 7: 41.

Birkett, M., J. Melville, P. Janulis, G. Phillips, N. Contractor, and B. Hogan. 2021. "Network Canvas: Key Decisions in the Design of an Interviewer-Assisted Network Data Collection Software Suite." *Social Networks* 66 (July): 114–24. https://doi.org/10.1016/j.socnet.2021.02.003.

Borgatti, Stephen P, Martin G Everett, Jeffrey C Johnson, and Filip Agneessens. 2024. *Analyzing Social Networks*. SAGE publications Ltd.

Borgatti, Stephen P., and Daniel S. Halgin. 2011. "On Network Theory." *Organization Science* 22 (5): 1168–81.

Borgatti, Stephen P., and Virginie Lopez-Kidwell. 2011. "Network Theory." In *Sage Handbook of Social*

Network Analysis, edited by John Scott and Peter Carrington. SAGE.

Brady, Sonya S., Linda Brubaker, Cynthia S. Fok, et al. 2020. "Development of Conceptual Models to Guide Public Health Research, Practice, and Policy: Synthesizing Traditional and Contemporary Paradigms." *Health Promotion Practice* 21 (4): 510–24. https://doi.org/10.1177/1524839919890869.

Collins, Caitlyn, Megan Tobias Neely, and Shamus Khan. 2024. "'Which Cases Do I Need?' Constructing Cases and Observations in Qualitative Research." *Annual Review of Sociology* 50 (1): 21–40. https://doi.org/10.1146/annurev-soc-031021-035000.

Dominguez, Silvia, and Betina Hollstein, eds. 2014. *Mixed Methods Social Networks Research: Design and Applications*. Structural Analysis in the Social Sciences 36. Cambridge University Press.

Doreian, P., V. Batagelj, and A. Ferligoj. 2005. *Generalized Blockmodeling*. Structural Analysis in the Social Sciences. Cambridge University Press.

Elwert, Felix. 2013. "Graphical Causal Models." In *Handbook of Causal Analysis for Social Research*. Springer.

Fox, John, and Robert Andersen. 2006. "Effect Displays for Multinomial and Proportional-Odds Logit Models." *Sociological Methodology* 36 (1): 225–55. https://doi.org/10.1111/j.1467-9531.2006.00180.x.

Frank, Ove. 1981. "A Survey of Statistical Methods for Graph Analysis." *Sociological Methodology* 12: 110–55. https://doi.org/10.2307/270740.

Freeman, Linton C. 2000. "Visualizing Social Networks." *Journal of Social Structure* 1 (1).

Freeman, Linton C. 2004. *The Development of Social Network Analysis: A Study in the Sociology of Science*. Empirical Press.

Halpern, Orit. 2015. *Beautiful Data: A History of Vision and Reason Since 1945*. Duke University Press.

Healy, Kieran. 2017. "Fuck Nuance." *Sociological Theory* 35 (2): 118–27. https://doi.org/10.1177/0735275117709046.

Healy, Kieran. 2018. *Data Visualization: A Practical Introduction*. Princeton University Press.

Healy, Kieran, and James Moody. 2014. "Data Visualization in Sociology." *Annual Review of Sociology* 40 (1): 105–28. https://doi.org/10.1146/annurev-soc-071312-145551.

Hunter, David R., Mark S. Handcock, Carter T. Butts, Steven M. Goodreau, and Martina Morris. 2008. "Ergm: A Package to Fit, Simulate and Diagnose Exponential-Family Models for Networks." *Journal of Statistical Software* 24 (3). https://doi.org/10.18637/jss.v024.i03.

Kadushin, Charles. 2012. *Understanding Social Network Analysis: Theories, Concepts & Findings*. Oxford University Press.

Krivitsky, Pavel N., David R. Hunter, Martina Morris, and Chad Klumb. 2023. "Ergm 4: New Features for Analyzing Exponential-Family Random Graph Models." *Journal of Statistical Software* 105 (6). https://doi.org/10.18637/jss.v105.i06.

Krivitsky, Pavel N., Martina Morris, Mark S. Handcock, et al. 2026. *Exponential Random Graph Models (ERGMs) Using Statnet*. Https://statnet.org/workshop-ergm/ergm_tutorial.html.

Leifer, Eric M. 1992. "Denying the Data: Learning from the Accomplished Sciences." *Sociological Forum* 7 (2): 283–99. https://doi.org/10.1007/BF01125044.

Lena, J. C. 2006. "Social Context and Musical Content of Rap Music, 1979-1995." *Social Forces* 85 (1): 479–95. https://doi.org/10.1353/sof.2006.0131.

Lena, Jennifer C. 2004. "Meaning and Membership: Samples in Rap Music, 1979–1995." *Poetics* 32 (3-4): 297–310. https://doi.org/10.1016/j.poetic.2004.05.006.

Lena, Jennifer C., and Mark C. Pachucki. 2013. "The Sincerest Form of Flattery: Innovation, Repetition, and Status in an Art Movement." *Poetics* 41 (3): 236–64. https://doi.org/10.1016/j.poetic.2013.02.002.

Lind, Benjamin. 2012. "Lessons on Exponential Random Graph Modeling from Grey's Anatomy Hook-Ups." In *Bad Hessian*. Https://badhessian.org/2012/09/lessons-on-exponential-random-graph-modeling-from-greys-anatomy-hook-ups/.

Liu, Wenlin, Anupreet Sidhu, Amanda M. Beacom, and Thomas W. Valente. 2017. "Social Network Theory." In *The International Encyclopedia of Media Effects*, 1st ed., edited by Patrick Rössler, Cynthia A. Hoffner, and Liesbet Zoonen. Wiley. https://doi.org/10.1002/9781118783764.wbieme0092.

Martin, John Levi. 2018. *Thinking Through Statistics*. University of Chicago Press.

McCarty, Christopher, Miranda J. Lubbers, Raffaele Vacca, and José Luis Molina. 2019. *Conducting Personal Network Research: A Practical Guide*. Methodology in the Social Sciences Series. Guilford Publications.

Miguel, Edward, Garret Christensen, and Jeremy Freese. 2019. *Transparent and Reproducible Social Science Research: How to Do Open Science*. University of California Press.

Moody, James, Daniel A. McFarland, and Skye Bender-DeMoll. 2005. "Dynamic Network Visualization." *American Journal of Sociology* 110 (4): 1206–41.

Moody, James, and Peter J. Mucha. 2013. "Portrait of Political Party Polarization." *Network Science* 1 (1): 119–21. https://doi.org/10.1017/nws.2012.3.

Narayanan, Arvind, and Vitaly Shmatikov. 2009. "De-Anonymizing Social Networks." *2009 30th IEEE Symposium on Security and Privacy*, May, 173–87. https://doi.org/10.1109/SP.2009.22.

Neal, Zachary P., Zack W. Almquist, James Bagrow, et al. 2024. "Recommendations for Sharing Network Data and Materials." *Network Science* 12 (4): 404–17. https://doi.org/10.1017/nws.2024.16.

Newman, Mark E., and M. Girvan. 2004. "Finding and Evaluating Community Structure in Networks." *Physical Review E.* 69: no.026113.

Ognyanova, Katherine. 2025. *Network Visualization with R*. Www.kateto.net/network-visualization.

Ortiz Ruiz, Francisca, Paola Tubaro, José-Luis Molina, et al. 2026. "Recommendations for Conducting Ethical Network Research." *Network Science* 14 (e17): 1–11.

Pache, Anne-Claire, and Filipe Santos. 2010. "When Worlds Collide: The Internal Dynamics of Organizational Responses to Conflicting Institutional Demands." *Academy of Management Review* 35 (3): 455–76.

Partha, Dasgupta, and Paul A David. 1994. "Toward a New Economics of Science." *Research Policy* 23 (5): 487–521.

Pasquale, Dana K., Tom Wolff, Gabriel Varela, et al. 2025. "Considerations for Social Networks and Health Data Sharing: An Overview." *Annals of Epidemiology* 102 (February): 28–35. https://doi.org/10.1016/j.annepidem.2024.12.014.

Potterat, J. J., D. E. Woodhouse, S. Q. Muth, et al. 2004. "Network Dynamism: History and Lessons of the Colorado Springs Study." In *Network Epidemiology: A Handbook for Survey Design and Data Collection*, edited by Martina Morris. Oxford University Press.

Ragin, Charles C., and Howard S. Becker. 1992. *What Is a Case?: Exploring the Foundations of Social Inquiry*. Cambridge University Press.

Rawlings, Craig M., Jeffrey A. Smith, James Moody, and Daniel A. McFarland, eds. 2023. "How Are Social Network Data Visualized?" In *Network Analysis: Integrating Social Network Theory, Method, and Application with R*. Structural Analysis in the Social Sciences. Cambridge University Press. https://doi.org/10.1017/9781139794985.006.

Rocher, Luc, Julien M. Hendrickx, and Yves-Alexandre de Montjoye. 2019. "Estimating the Success of Re-Identifications in Incomplete Datasets Using Generative Models." *Nature Communications* 10 (1): 3069. https://doi.org/10.1038/s41467-019-10933-3.

Sayama, Hiroki, Catherine Cramer, Mason A. Porter, Lori Sheetz, and Stephen Uzzo. 2016. "What Are Essential Concepts about Networks?" *Journal of Complex Networks* 4 (3): 457–74. https://doi.org/10.1093/comnet/cnv028.

Schwabish, Jonathan. 2021. *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks*. Columbia University Press.

Schwarze, Alice C., and Mason A. Porter. 2021. "Motifs for Processes on Networks." *SIAM Journal on Applied Dynamical Systems* 20 (4): 2516–57. https://doi.org/10.1137/20M1361602.

Snijders, Tom A. B., and Christian Steglich. 2026. *Stochastic Actor-Oriented Models for Longitudinal Networks*. Structural Analysis in the Social Sciences. Cambridge University Press.

Sweller, John. 1988. "Cognitive Load During Problem Solving: Effects on Learning." *Cognitive Science* 12 (2): 257–85. https://doi.org/10.1016/0364-0213(88)90023-7.

Timmermans, Stefan, and Iddo Tavory. 2022. *Data Analysis in Qualitative Research: Theorizing with Abductive Analysis*. The University of Chicago Press.

Tufte, Edward R. 2006. *Beautiful Evidence*. Vol. 1. Graphics Press Cheshire, CT.

Tufte, Edward R. 1983. *The Visual Display of Quantitative Information*. Graphics Press.

Vacca, Raffaele. 2020. "Structure in Personal Networks: Constructing and Comparing Typologies." *Network Science* 8 (2): 142–67. https://doi.org/10.1017/nws.2019.29.

Vertesi, Janet. 2025. "Theory-Laden Data Visualization, Drawing-As and Seeing-As in Sociology and in Data Science." *The American Sociologist* 56 (June): 440–54. https://doi.org/10.1007/s12108-025-09658-2.

Watts, Duncan J, and Steven H Strogatz. 1998. "Collective Dynamics of 'Small-World' Networks." *Nature* 393: 440–42.

Wickham, Hadley, Mine Cetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. 2nd ed. O'Reilly.

Wolff, Tom, Gabriel Varela, and Jonathan Eastwood. 2026. "Building Computational Literacy in Undergraduate Sociology Using Software: An Example in Teaching Social Network Analysis." *Teaching Sociology*, ahead of print. https://doi.org/10.1177/0092055X261422479.

Zhao, Jun, and Christabel L. Rogalin. 2024. "Discrimination in Sentencing: Showing Remorse and the Intersection of Race and Gender." *Social Psychology Quarterly* 87 (4): 309–32. https://doi.org/10.1177/01902725231219690.

Zimmer, Michael. 2010. "'But the Data Is Already Public': On the Ethics of Research in Facebook." *Ethics and Information Technology* 12 (4): 313–25. https://doi.org/10.1007/s10676-010-9227-5.

Zinsser, William. 1993. *Writing to Learn*. Harper Collins.